Maria Pilligua | Excellence Fellow EPFL

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Computer vision researcher across image, video, geometry, and motion. Currently building a real-time text-to-motion model with kinematic conditioning at EPFL, with prior work in inverse rendering, neural video editing, and diffusion-based generation. 5 papers at CVPR/ECCV and a shipped AI feature at GoodNotes used by millions.

Education

École Polytechnique Fédérale de Lausanne (EPFL) – MSc in Data Science Sep 2025 – Aug 2027
- **Excellence Fellowship (ESOP)** awarded to top 0.004% M.Sc. students across EPFL. **GPA**: 5.4/6

Autonomous University of Barcelona (UAB) – BSc in Artificial Intelligence Sep 2021 – Jun 2025
- **GPA**: 9.41/10 | Best grade over all engineering undergraduate degrees | 3 merit scholarships.

Experience

Student Researcher - EPFL Computer Vision Lab (Lausanne, CH) Feb 2025 - Present
- Investigating transformer attention mechanisms in VGGT for multiview 3D scene reconstruction, researching alternatives that make the model faster and less computationally expensive for real-time use.

Machine Learning Intern - GoodNotes (London, UK) Jul 2025 - Sep 2025
- Proposed and led a diffusion-based pipeline that generates editable vector graphics (SVG) from text and ships into the GoodNotes drawing canvas. Output stays fully editable inside the app, not flattened to pixels.
- Prototyped 10+ model variants across prompting, filtering, ranking, and vectorization. Reduced inference latency from ∼40s to ∼7s while improving quality, deployed to millions of users.

Student Researcher - Computer Vision Center (Barcelona, Spain) Jan 2023 - Jul 2025
- Research on image, video, and 3D generative modeling: transformer-, diffusion-, hypernetwork-, and CNN-based models for neural video editing, controllable relighting, document restoration, low-light enhancement, and continual learning.
- Authored and presented multiple papers at major CV conferences and workshops (CVPR 2025, ECCV 2024). Awarded four competitive research scholarships for academic excellence.

AI Consultant - IDISC Information Technology (Remote) Nov 2024 - Apr 2025
- Evaluated and prototyped AI pipelines for document digitization from images using OCR, layout analysis, image enhancement, and LLMs for text understanding.

Selected Projects

Controllable Autoregressive Text-to-Motion Generation | EPFL CS-503
- Real-time text-to-motion model that lets the user steer generation mid-sequence: at any frame, specify a full-body pose, hand or foot position, or a path the character must follow. Diffusion approaches can't do this online; ours generates autoregressively.
- T2M-GPT with a VQ-VAE motion codebook and 18-layer causal Transformer (1024 hidden dim, 16 heads), trained on BONES (64k motion-capture sequences). R@3 30.6 and FID 0.449 on the Kimodo Motion Generation benchmark.
- Two constraint-injection architectures: a ControlNet-style cross-attention adapter with a soft-codebook differentiable trajectory loss, and a prefix-token approach with a custom causal mask, enabling online steering at sub-diffusion latencies.

Controllable Relighting | Bachelor Thesis
- Developed a Vision Transformer-based inverse rendering model that disentangles geometry, material, and illumination from single images, enabling controllable relighting of scenes.
- Explored conditioning strategies with diffusion models (Stable Diffusion, ControlNet, IP-Adapter) to preserve scene structure while allowing user-controlled lighting edits.

HyperNVD: Accelerated Video Editing with Hypernetworks | CVPR 2025 https://hypernvd.github.io/
- Proposed a hypernetwork-based framework to accelerate the fitting of Neural Implicit Representations for video editing, reducing per-scene optimization from 2 hours to 20 minutes (6x speedup).
- Extends state-of-the-art work in video decomposition (e.g., Adobe's NVD) by introducing a meta-learning prior that enables near-instant adaptation to novel video content.

Awards

- **EPFL MSc Excellence Fellowship (ESOP)** (2025): top 0.004% (30 students) of incoming MSc students; 40k CHF grant.
- **Best Research Project, UAB Computer Vision Center** (2024 and 2025): awarded to 1 student per year.
- **Best Academic Record across all UAB engineering degrees** (2025): granted to 4 students.
- **Best Academic Record, Ibiza** (2025): granted to 2 students across all degrees in the region.
- **CVPR & ECCV DEI Scholarships** (2024–2025): 29 of 524 applicants selected for ECCV.
- **4 research scholarships from the Computer Vision Center** (2023–2025): competitive grants awarded to pursue research.
- **Honors in 18 subjects** (2021–2025): including 3D Vision, Generative Models, Deep Learning, CV, and NLP.

Publications

[1] **Maria Pilligua**, D. Xue, J. Vázquez-Corral. *"HyperNVD: Accelerating Neural Video Decomposition via Hypernets."* **CVPR 2025**.

[2] **Maria Pilligua**, D. Serrano-Lozano, P. Peng, R. Baldrich, M. S. Brown, J. Vázquez-Corral. *"Evaluating Low-Light Image Enhancement Across Multiple Intensity Levels."* **CVPR Workshop (Findings) 2026**.

[3] **Maria Pilligua**, N. De La Fuente, D. Vidal, A. Soutif, A. Barskey. *"PAH: Prototype Augmented Hypernetworks."* **CVPR Workshop (LatinX in Computer Vision) 2025**.

[4] **Maria Pilligua**, N. Biescas, J. Vázquez-Corral, J. Lladós, E. Valveny, S. Biswas. *"LayeredDoc: Domain Adaptive Document Restoration with a Layer Separation Approach."* **ECCV Workshop (WiCV) 2025**.

[5] D. Serrano-L., F. A. Molina B., D. Xue, Y. Yang, **Maria Pilligua**, R. Baldrich, M. Vanrell, J. Vázquez-Corral. *"PromptNorm: Image Geometry Guides Ambient Light Normalization."* **CVPR Workshop (NTIRE) 2025**.

Skills

Programming Languages: Python, C++, Bash

Specialized Knowledge: Generative AI (diffusion, autoregressive), neural rendering & inverse rendering, multiview 3D reconstruction, motion synthesis, video editing, transformers, vision-language models (CLIP), VQ-VAE, hypernetworks, real-time inference.

Tools: PyTorch, Lightning AI, OpenCV, Hugging Face, wandb, Git, Docker, AWS (S3, EC2), LaTeX, SLURM.

Languages: Spanish (native), English (C1), Catalan (C1)

Hackathons

MAPA: Multimodal agent controlling a quadruped robot (voice, VLM, SLAM) – RoboHack EPFL 3rd Place 2026

- Built a tool-using agent to control a quadruped as a guide dog: app-based voice requests, speech output, Claude as VLM for scene understanding, and YOLO + SLAM for tracking, detection, and obstacle avoidance.

Flowchart detection – GoodNotes Hackathon 2nd Place 2025

- Designed a flowchart detection system for the GoodNotes app using a combination of YOLO-based layout analysis and custom heuristics enabling users to easily convert hand-drawn flowcharts into editable digital formats.

TRAIN TALES – UAB The Hack (Interactive AI Travel Companion for Kids) 2nd Place 2025

- Developed an interactive web application that narrates location-aware stories during train journeys and generates quizzes to engage children, particularly supporting kids with hyperactivity.

INDITECH Challenge – HackUPC (Multimodal AI for E-Commerce) 2nd Place 2023

- Developed a voice- and gesture-controlled fashion shopping app using Whisper, Fashion-CLIP, YOLO, and Hugging Face transformers, with a Flask-based backend and real-time interaction.